Chemical Accuracy at Scale

Background

Density functional theory (DFT) is the workhorse for computing the properties of molecules and materials: formation and reaction energies, structures, band gaps, phonons, magnetism and dynamics. For the ground-state energy and density it is exact in principle: the Hohenberg–Kohn theorems guarantee an exact functional exists, and Kohn–Sham theory makes it computable. But the Kohn–Sham construction is only exact for ground-state quantities. The Kohn–Sham eigenvalue gap is *not* the fundamental band gap even with the exact functional: the two differ by the derivative discontinuity of the XC energy, which no smooth semilocal functional can produce. Excited states, quasiparticle energies and response properties likewise lie outside ground-state DFT and need its generalizations (GKS, TDDFT, GW). Three limits follow:

- **Accuracy.** All of the unknown physics is packed into the exchange–correlation (XC) functional. Today's functionals are typically off by 2–10 kcal/mol on energies and fail qualitatively for strong correlation and charge transfer. Chemical accuracy (≈1 kcal/mol) still requires methods like CCSD(T), which scale as $O(N^7)$.
- **Scope.** For quantities that are not ground-state properties, above all band gaps, a better functional is not enough: the theory itself has to be extended, and the extensions (generalized Kohn–Sham, TDDFT, relativistic and superconducting DFT) each bring their own unknown functionals and unresolved formal questions.
- **Cost.** Standard Kohn–Sham DFT scales as $O(N^3)$, and hybrid functionals are worse. That caps routine calculations at roughly 10^3 atoms, well below what interfaces, disordered materials and devices need.

These problems test whether AI systems can make real progress on any of these fronts. Each task is posed so that its answers can be **scored automatically and re-run as models improve**.

Tasks

1. Efficiency. Find approximations to DFT that scale linearly with system size without losing chemical accuracy. Implement this in [code TBD]. *Scored by:* measured scaling exponent, and error versus the standard $O(N^3)$ calculation (target ≤ 1 kcal/mol) on a fixed suite of insulators, semiconductors and metals.

2. Accuracy. Construct a new XC functional that outperforms [reference functional] on [properties] across [benchmark set], for reasons that generalize: by discovering and enforcing exact constraints that existing functionals violate, by introducing new physically motivated forms or ingredients, or by machine-learned constructions that satisfy exact conditions by construction. Re-fitting existing functional forms, or any parameter tuned to the benchmark, does not count. *Scored by:* error on hidden test sets against CCSD(T), QMC or experimental reference data, including property classes absent from the development set, with no property class allowed to get worse than a set tolerance; plus a reviewer's judgment that the construction is new.

3. Theory. - (a) Find tighter bounds or higher-order expansions of known properties of the exact functional (e.g. the Lieb–Oxford bound, the gradient expansion, scaling relations). - (b) Find new exact sum rules the exact XC functional should satisfy, and show that enforcing them gives better ground-state properties.

Scored by: proof checking (formal or expert review), numerical tests against exact or near-exact densities, and improvement on the Task 2 benchmark when the new constraint is enforced.

4. Extensions. Extend the above results to generalizations of DFT, such as generalized Kohn–Sham (GKS), real-time and linear-response TDDFT, relativistic DFT and superconducting DFT. *Scored by:* the analogous accuracy or cost metrics for that generalization (e.g. excitation energies, spin–orbit splittings, superconducting T_c).

Benchmarking

- **Fixed harness:** standard inputs, reference data and scoring scripts, so results stay comparable over time.
- **Hidden test set:** refreshed periodically to avoid contamination.
- **Leaderboard per task:** tracks the best AI result against the best human/literature baseline (e.g. r²SCAN, ωB97M-V, existing linear-scaling codes).

Open decisions

- Target code for Task 1
- Reference functional, properties and benchmark set for Task 2
- How Task 3 proofs are verified